Olga Seleznova

Data Scientist

Toronto, ON | 343-322-1181 | olgaselesnyova@gmail.com |
linkedin.com/in/olga-seleznova | github.com/OlgaSeleznova

Data Scientist with over 3 years of experience in Deep Learning, Natural Language Processing (NLP), Large Language Modeling (LLMs), and Speech Recognition. Proficient in model debugging and optimization, and competency in translating complex data into actionable insights. Advanced degree in Data Science, complemented by AI and Machine Learning technologies certifications.

CORE COMPETENCIES

- Experience with end-to-end machine learning modeling, using Pytorch, Transformer-based architectures, and Large Language Models (LLM).
- Managing large-scale structured and unstructured data using Python, SQL, and MongoDB.
- Leveraging AWS, Docker, and Git to manage machine learning lifecycles, ensuring scalable and efficient cloud-based deployments.
- Competent in interpreting and communicating complex data findings to business stakeholders and delivering meaningful presentations.
- Experienced in fostering projects within cross-functional teams, with a proactive approach to problem-solving.

TECHNICAL EXPERTISE

Programming languages: Python, SQL, MongoDB, Bash
Machine learning tools: NumPy, Pandas, Scikit-learn, Matplotlib, and others
Machine learning algorithms: XGBoost, Random Forest, time series (ARIMA), Logistic regression, Linear regression, SVM, HDBSCAN, Kmeans and others
Deep learning modeling: PyTorch, Transformer architectures, Hugging Face, Weights & Biases
Natural language processing Tools: Word2Vec, SpaCy, BERT, vector similarity (FAISS)
Speech Recognition packages: Whisper, Torch audio, Wave2vec, HuBERT, Librosa
Large Language Models (LLM), AI tools and methodologies: Prompt engineering, Ollama, Llama, Mistral, OpenAI, LangChain, LlamaIndex, Retrieval Augmented Generation (RAG), GitHub copilot
Version control: Git, GitHub
Cloud technologies: AWS, Docker
IDEs: VSCode, PyCharm, Jupyter Notebooks, Google Colab
UI frameworks for Machine learning: Gradio, Streamlit

PROFESSIONAL EXPERIENCE

Research Engineer March 2023 – June 2024
Technion - Israel Institute of Technology, Israel

- Operated within the 'Speech, Language, and Deep Learning' Lab, driving the modeling and research of deep learning projects in the speech and language domains, utilizing Python, PyTorch, Whisper, Torchaudio, NumPy, Matplotlib, and other advanced technologies.
- Developed over 4 clean and reusable packages for academic purposes by creating templates and streamlining submission processes.
- Conducted research in automatic speech recognition using Whisper, HuBERT, and wav2vec libraries and performed speech alignment with Montreal Forced Aligner and restoration with UMAP model.
- Managed virtual machines and created server and Docker environments to support deep learning projects, ensuring a functional and scalable infrastructure.

Data Scientist September 2021 – September 2022
Skai (Software Development), Israel

- Provided actionable insights from large datasets of unstructured reviews to management, facilitating data-driven decision-making and enhancing strategic product development based on customer feedback.

- Led the research, development, and implementation of advanced NLP systems, including the fine-tuning and error analysis of BERT models for text classification and T5 for question-answering tasks.
- Performed data exploration, data visualization, feature engineering, and modeling using classical ML algorithms like XGBoost, Random Forest, time series (ARIMA), Logistic regression, and Linear regression, to deliver data stories that could lead to actionable insights.
- Designed and deployed semantic similarity and keyword extraction modules to advance insights mining, improving the accuracy and relevance of extracted information.
- Coordinated a multidisciplinary team of Python and R developers, data analysts, and statisticians under a product manager's guidance to achieve project deliverables and ensure compliance with technical standards.

Data Science Intern February 2021 – July 2021

TRG Solutions (IT Services and IT Consulting), Israel

- Fine-tuned the DistilBERT model from HuggingFace, increasing sentiment classification accuracy from 47% to 86.20% across three categories (positive, negative, and neutral).
- Implemented text classification models, including Random Forest, CatBoost, XGBoost, RoBERTa, and DistilBERT, utilizing Scikit-learn and Transformers packages to enhance predictive accuracy and performance.
- Delivered a comprehensive presentation of findings to students and lecturers, communicating technical details and insights.
- Collaborated closely with team members and operated under the guidance of a senior data scientist, efficiently sharing responsibilities and contributing to team success.

Researcher August 2020 – December 2020

University of Haifa, Israel

- Contributed to a published paper on informativeness versus emotionality in medical forums, identifying key features to differentiate between these concepts within the Integrative Pain (iPain) laboratory.
- Achieved approximately 85% accuracy in classifying Hebrew posts from medical forums, addressing low-scale, unbalanced data challenges.
- Conducted comprehensive EDA, text preprocessing, and classification using Python libraries, including Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, and PyTorch transformers.

EDUCATION AND PROFESSIONAL DEVELOPMENT

Master's Degree in Data Science 2022

University of Haifa, Israel

Advancement Data Science Program 2021

Ydata, Intensive one-year career advancement program in Data Science, Israel

CERTIFICATIONS

Generative AI with Large Language / DeepLearning AI - Coursera 2023

Introduction to Machine Learning in Production / DeepLearning AI - Coursera 2023

Mathematics for Machine Learning: Multivariate Calculus and Linear Algebra / Coursera 2020

Data Structures and Algorithms in Python / Udacity 2020

Using Python to Access Web Data / University of Michigan – Coursera 2018

EXTERNAL PROJECTS

VoyageVocab Project: Language Learning Assistant for Travellers

Developed an application utilizing Llama and LangChain to generate travel-related phrases for tasks such as ordering food or buying tickets, with a user-friendly interface implemented through the Gradio package.

Link: github.com/OlgaSeleznova/VoyageVocab

Job Posting Relevance

An AI-powered tool designed to determine how relevant a job posting is to a candidate's resume. The system leverages the RAG methodology and agentic flow, based on the Langchain framework to analyze how well a resume aligns with the specific requirements of a job posting. Data was managed using MongoDB and user experience with Streamlit application.

Link: github.com/OlgaSeleznova/JobPostingRelevancy